ECS 1100

Introduction to Engineering and Computer Science

WHILE YOU GET SETTLED

Sit anywhere — but not next to someone you already know.

You will be working with the people around you today.

What This Course Is

One credit hour, once a week, and it is MORE than a survey of majors.

What most first-year courses do

One major a week. Mechanical this week, electrical next.

You cannot choose well between things you have only seen described one at a time.

What this course does

Seven SECTORS — AI, semiconductors, health, energy, manufacturing, consulting, cybersecurity.

In each one you map where all TEN ECS degree programs act.

Ten degree programs · five research thrusts · six cross-cutting technologies

Nobody memorizes that today. You will meet it seven times, one sector at a time, with someone in the room who does the work.

Three things today: your concept map, your Connections practice, and the Engagement Requirement.

You are also in ECS 1210 with the same people. That course has the coffee grinder. This one has the sectors.

1

Concept Maps

The tool you will use nine times this semester.

How to Build a Concept Map

We will use this learning tool all semester!

This is NOT a mind map

A mind map is bubbles connected by lines. It shows that things are related.

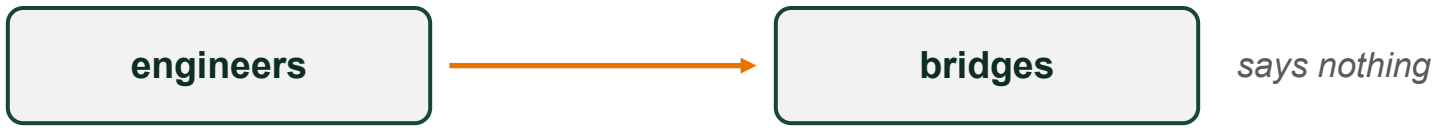
It does not say HOW they are related, so it cannot be right or wrong.

A concept map makes CLAIMS

Concepts in boxes. Every arrow carries a LABEL — a few words saying what one does to the other.

That label turns a picture into a set of statements you can argue with.

THE TEST: read any box → arrow → box out loud. Does it form a sentence that could be TRUE or FALSE?



says nothing

Why We Use Concept Maps

What the research says, and what it means for you.

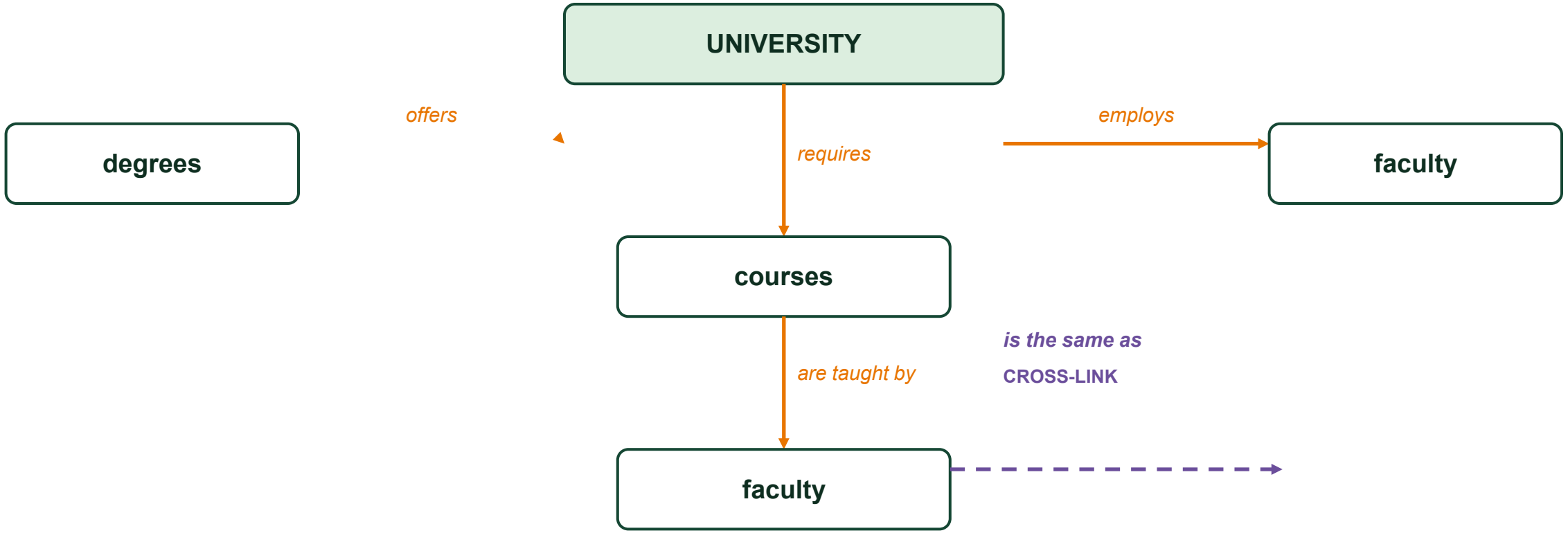
They show CONNECTIONS, not just terms	Concepts are held in relation to each other rather than as an isolated list. That is what makes knowledge usable.
Struggling to label a link is INFORMATION	If you cannot say what one thing does to another, you have found a gap in your own understanding. That is the most useful thing a map does.
Engineering is connection work	Knowing a concept is not enough. You have to connect it to problems, constraints, users, and consequences.
Students consistently rate them highly	In published studies, 95 to 100 percent of engineering students said the concept map assignment helped them connect ideas across a course.

Nine maps this semester: one today, seven sector maps, and one in Week 16 that you compare against today's.

Let's Build One

0:15

Three minutes, on something everyone in this room knows.



Read it aloud: “University OFFERS degrees.” “Degrees REQUIRE courses.” “Courses ARE TAUGHT BY faculty.” Every one is a claim.

Your Baseline Map

Fifteen minutes. The first of nine you will build this semester.

Draw what you currently believe an engineer or computer scientist does.

Concepts in boxes	Aim for eight to fifteen. More is fine.
Every arrow labeled	If you cannot label it, you do not yet know what the relationship is. That is useful information.
General at the top	Specific below it.
At least one cross-link	Join two branches that are not directly above or below each other.

Make it HONEST, not good. You build this again in Week 16 and compare them — a thin map today is what makes December's map mean something.

2

Connections

Five structured conversations. We practice one now.

3

The Engagement Requirement

30% of your grade. One activity should be done over the next week.

The Engagement Requirement

0:53

30% of your grade. Five categories, five separate activities.

A	**ECS Welcome Event**	An ECS Welcome Week event, or the MECH Freshman Town Hall on Friday, August 28.
B	**Career Fair, Workshop, or Networking Session**	A career fair, a career workshop, a networking session, a resume review, or an approved Handshake profile.
C	**ECS Student Organization**	A general meeting, recruitment night, or project team open house for any ECS-affiliated organization.
D	**Community and Discipline**	A seminar, research info session, UTDesign expo, office hours visit, or an alumni panel.
E	**Campus Life**	A UTD-wide organization or campus event that is NOT ECS-affiliated — sports, campus-wide events.

These are SEPARATE from your Connections. An engagement activity cannot also count as a Connections conversation.

CATEGORY A IS HAPPENING RIGHT NOW

You WILL finish one of the five this week.

ECS Welcome Week

Running now. Events are on the ECS calendar.

`calendar.utdallas.edu/department/ecs/calendar`

MECH Freshman Town Hall

Friday, August 28.

Category A in one afternoon — the same day your first homework is due.

Go. Take a photo. Log it. Extra credit for a selfie with a faculty member.

Neither of these runs again in October. This is the easiest of your five activities and it is available for about two more weeks.

How This Course Runs

FRIDAY — engagement

Your Connections conversations and your Engagement Requirement activities.

Always a Friday.

BEFORE NEXT CLASS — academic

At most two items: follow-up from the session you just had, and prep for the next one.

Four things are graded: Connections 10% · Engagement Requirement 30% · In-class participation 30% · Homework 30%

Before next class

- **THIS WEEK** — Go to an ECS Welcome Week event or the MECH Town Hall on Friday. That is Category A done.
- **HOMEWORK 1** — Three Threes of the Syllabus, AND your Engagement Category Plan — what you plan to attend for each of the five categories, and your next step to get there.
- **On Canvas** — The Engagement Requirement page, and the full Connections logistics. Read both before Friday.
- **Next week** — Your resume, and the course survey. Bring a laptop, and think about anything you have built, fixed, or been paid to do.